PHB_221_Final_Project

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2025-11-13

Dataset and Libraries

```r
# libraries
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.3.3

## Warning: package 'purrr' was built under R version 4.3.3

## Warning: package 'lubridate' was built under R version 4.3.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
# dataset
data <- read_csv("NCHS_-_Leading_Causes_of_Death__United_States.csv")
```

```
## Rows: 10868 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): 113 Cause Name, Cause Name, State
## dbl (3): Year, Deaths, Age-adjusted Death Rate
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Data Collection and Cleaning

```r
# view first few rows of data
head(data)
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                  <dbl>
## 1  2017 Accidents (unintention~ Unintention~ Unit~ 169936                   49.4
## 2  2017 Accidents (unintention~ Unintention~ Alab~   2703                   53.8
```

```
## 3  2017 Accidents (unintention~ Unintention~ Alas~    436                    63.7
## 4  2017 Accidents (unintention~ Unintention~ Ariz~   4184                    56.2
## 5  2017 Accidents (unintention~ Unintention~ Arka~   1625                    51.8
## 6  2017 Accidents (unintention~ Unintention~ Cali~  13840                    33.2
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

```r
# check the number of rows and cols
dim(data)
```

```
## [1] 10868     6
```

```r
# filter data
filtered_data <- data[data$`Cause Name` == "Heart disease", ]

head(filtered_data)
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                  <dbl>
## 1  2017 Diseases of heart (I00~ Heart disea~ Unit~ 647457                   165
## 2  2017 Diseases of heart (I00~ Heart disea~ Alab~  13110                   223.
## 3  2017 Diseases of heart (I00~ Heart disea~ Alas~    814                   135
## 4  2017 Diseases of heart (I00~ Heart disea~ Ariz~  12398                   142.
## 5  2017 Diseases of heart (I00~ Heart disea~ Arka~   8270                   224.
## 6  2017 Diseases of heart (I00~ Heart disea~ Cali~  62797                   143.
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

```r
dim(filtered_data)
```

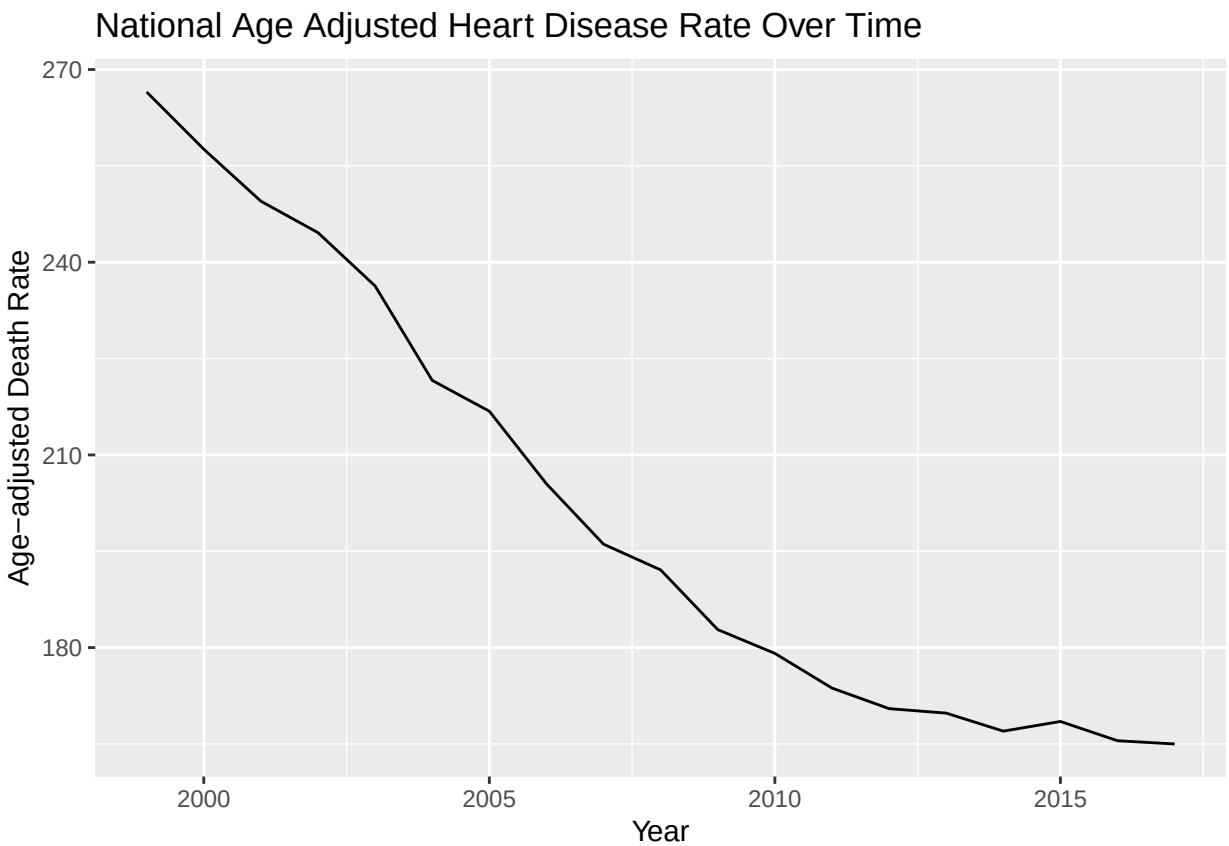
```
## [1] 988   6
```

Explantory Analysis

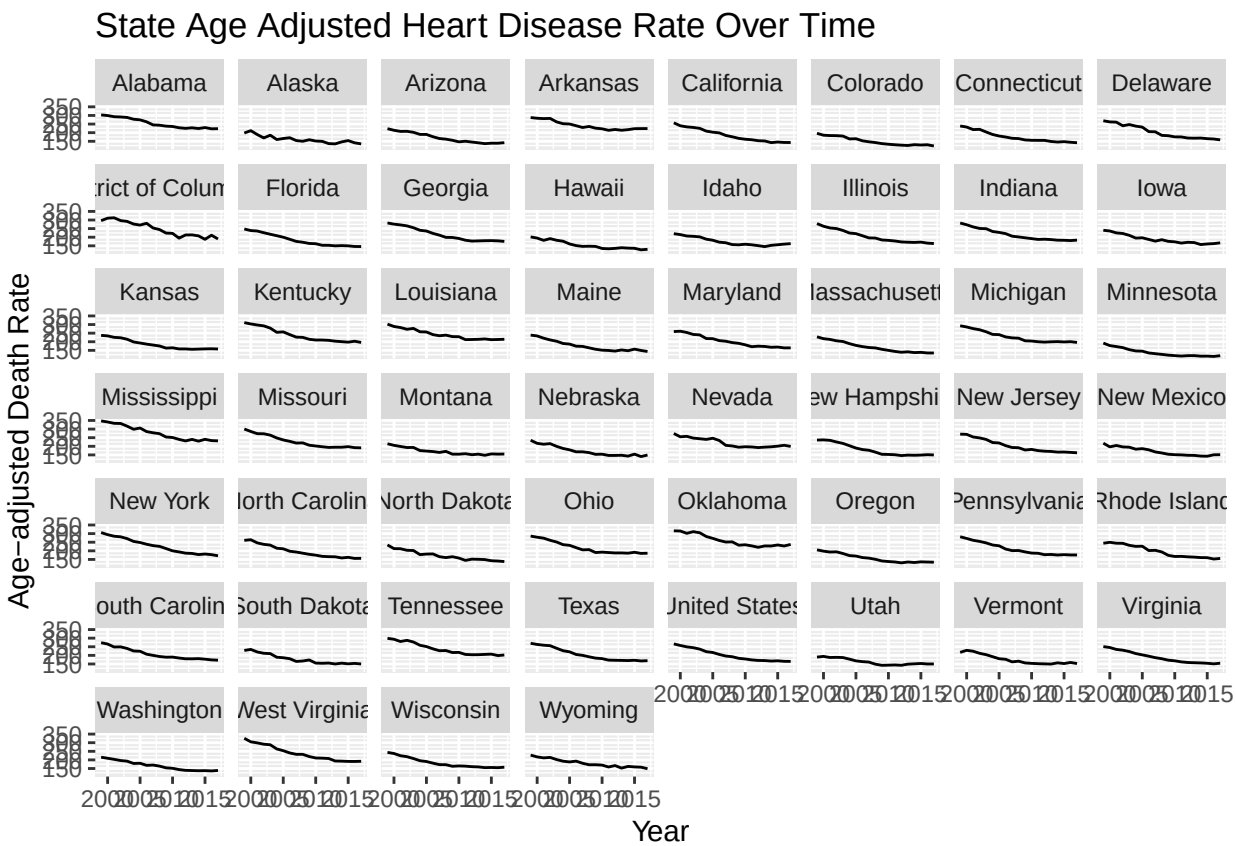
```r
# national level
national_data <- data |>
  filter(`Cause Name` == "Heart disease", State == "United States")

head(national_data)
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                  <dbl>
## 1  2017 Diseases of heart (I00~ Heart disea~ Unit~ 647457                   165
## 2  2016 Diseases of heart (I00~ Heart disea~ Unit~ 635260                   166.
## 3  2015 Diseases of heart (I00~ Heart disea~ Unit~ 633842                   168.
## 4  2014 Diseases of heart (I00~ Heart disea~ Unit~ 614348                   167
## 5  2013 Diseases of heart (I00~ Heart disea~ Unit~ 611105                   170.
## 6  2012 Diseases of heart (I00~ Heart disea~ Unit~ 599711                   170.
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

```r
ggplot(data = national_data, mapping = aes(x = Year, y = `Age-adjusted Death Rate`)) +
  geom_line() +
  labs(title = "National Age Adjusted Heart Disease Rate Over Time")
```




```r
# state level
ggplot(filtered_data, aes(x = Year, y = `Age-adjusted Death Rate`)) +
  geom_line() +
  facet_wrap(~ State) +
  labs(title = "State Age Adjusted Heart Disease Rate Over Time")
```



Statistical Testing

- notes:
  - anova, compares group means to the grand mean

```
# one way anova testing
state_only <- filtered_data[filtered_data$State != "United States", ]

head(state_only)
```

```
## # A tibble: 6 x 6
##    Year `113 Cause Name`        `Cause Name` State Deaths Age-adjusted Death R~1
##   <dbl> <chr>                   <chr>        <chr>  <dbl>                   <dbl>
## 1  2017 Diseases of heart (I00~ Heart disea~ Alab~  13110                    223.
## 2  2017 Diseases of heart (I00~ Heart disea~ Alas~    814                    135
## 3  2017 Diseases of heart (I00~ Heart disea~ Ariz~  12398                    142.
## 4  2017 Diseases of heart (I00~ Heart disea~ Arka~   8270                    224.
## 5  2017 Diseases of heart (I00~ Heart disea~ Cali~  62797                    143.
## 6  2017 Diseases of heart (I00~ Heart disea~ Colo~   7060                    123.
## # i abbreviated name: 1: `Age-adjusted Death Rate`
```

```
one_way <- aov(`Age-adjusted Death Rate` ~ State, data = state_only)
summary(one_way)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## State        50 957770   19155   18.05 <2e-16 ***
## Residuals   918 974201    1061
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# plot mean differences
```

Predictive Modeling

```r
# linear regression
m1 <- lm(`Age-adjusted Death Rate` ~ Year + State, data = state_only)
summary(m1)
```

```
##
## Call:
## lm(formula = `Age-adjusted Death Rate` ~ Year + State, data = state_only)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.708  -9.030  -1.771   8.062  38.040
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)               11064.1323   141.1943  78.361  < 2e-16 ***
## Year                         -5.3833     0.0703 -76.573  < 2e-16 ***
## StateAlaska                 -93.2421     3.8889 -23.976  < 2e-16 ***
## StateArizona                -84.2158     3.8889 -21.655  < 2e-16 ***
## StateArkansas               -12.8105     3.8889  -3.294  0.00103 **
## StateCalifornia             -68.9789     3.8889 -17.737  < 2e-16 ***
## StateColorado              -103.8263     3.8889 -26.698  < 2e-16 ***
## StateConnecticut            -78.5053     3.8889 -20.187  < 2e-16 ***
## StateDelaware               -50.6947     3.8889 -13.036  < 2e-16 ***
## StateDistrict of Columbia    -7.1895     3.8889  -1.849  0.06482 .
## StateFlorida                -71.0790     3.8889 -18.277  < 2e-16 ***
## StateGeorgia                -38.3737     3.8889  -9.867  < 2e-16 ***
## StateHawaii                 -99.0158     3.8889 -25.461  < 2e-16 ***
## StateIdaho                  -79.9000     3.8889 -20.545  < 2e-16 ***
## StateIllinois               -49.8053     3.8889 -12.807  < 2e-16 ***
## StateIndiana                -39.5316     3.8889 -10.165  < 2e-16 ***
## StateIowa                   -66.1105     3.8889 -17.000  < 2e-16 ***
## StateKansas                 -70.1158     3.8889 -18.030  < 2e-16 ***
## StateKentucky               -15.9000     3.8889  -4.089 4.72e-05 ***
## StateLouisiana              -11.9263     3.8889  -3.067  0.00223 **
## StateMaine                  -78.2053     3.8889 -20.110  < 2e-16 ***
## StateMaryland               -51.5632     3.8889 -13.259  < 2e-16 ***
## StateMassachusetts          -85.3632     3.8889 -21.950  < 2e-16 ***
## StateMichigan               -25.9158     3.8889  -6.664 4.59e-11 ***
## StateMinnesota             -117.0368     3.8889 -30.095  < 2e-16 ***
## StateMississippi             22.6842     3.8889   5.833 7.54e-09 ***
## StateMissouri               -27.1053     3.8889  -6.970 6.06e-12 ***
## StateMontana                -83.8684     3.8889 -21.566  < 2e-16 ***
## StateNebraska               -81.7474     3.8889 -21.020  < 2e-16 ***
## StateNevada                 -34.1263     3.8889  -8.775  < 2e-16 ***
## StateNew Hampshire          -74.6158     3.8889 -19.187  < 2e-16 ***
## StateNew Jersey             -51.8737     3.8889 -13.339  < 2e-16 ***
## StateNew Mexico             -84.6737     3.8889 -21.773  < 2e-16 ***
```

```
## StateNew York              -26.9947     3.8889  -6.941 7.34e-12 ***
## StateNorth Carolina        -57.9421     3.8889 -14.899  < 2e-16 ***
## StateNorth Dakota          -82.1737     3.8889 -21.130  < 2e-16 ***
## StateOhio                  -36.1842     3.8889  -9.304  < 2e-16 ***
## StateOklahoma                5.5895     3.8889   1.437  0.15098
## StateOregon                -96.3421     3.8889 -24.773  < 2e-16 ***
## StatePennsylvania          -42.7947     3.8889 -11.004  < 2e-16 ***
## StateRhode Island          -58.0895     3.8889 -14.937  < 2e-16 ***
## StateSouth Carolina        -46.2526     3.8889 -11.893  < 2e-16 ***
## StateSouth Dakota          -75.9895     3.8889 -19.540  < 2e-16 ***
## StateTennessee             -17.4895     3.8889  -4.497 7.77e-06 ***
## StateTexas                 -49.5737     3.8889 -12.747  < 2e-16 ***
## StateUtah                  -91.6842     3.8889 -23.576  < 2e-16 ***
## StateVermont               -78.2579     3.8889 -20.123  < 2e-16 ***
## StateVirginia              -65.3421     3.8889 -16.802  < 2e-16 ***
## StateWashington            -88.3105     3.8889 -22.708  < 2e-16 ***
## StateWest Virginia         -15.4368     3.8889  -3.969 7.77e-05 ***
## StateWisconsin             -71.3105     3.8889 -18.337  < 2e-16 ***
## StateWyoming               -72.9526     3.8889 -18.759  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.99 on 917 degrees of freedom
## Multiple R-squared:  0.9318, Adjusted R-squared:  0.928
## F-statistic: 245.7 on 51 and 917 DF,  p-value: < 2.2e-16
```

```r
# multiple regression


# fixed state effects
```

Evaluation and Interpretation

```r
# check models accuracies

# confusion matrix

# k means clustering
```